Project: Worked examples
Title: Lifting lug and pin joint
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A pinned lug between two cheek plates under a load along the lug: tension on the net section through the hole, bearing of the pin on the hole, shear tear-out beyond the hole, and the pin in double shear and in bending, each against an allowable stress; the largest utilisation governs.

WHAT IT CALCULATES	RELATION
Tension across the net section beside the hole	$\sigma_t = P/((W - d_h)\cdot t)$
Bearing of the pin on the lug	$\sigma_{br} = P/(d_p\cdot t)$
Tear-out shear beyond the hole, on two planes	$\tau_{to} = P/(2\cdot a\cdot t)$
Pin shear, in double shear	$\tau_p = P/(2\cdot\pi\cdot d_p^2/4)$
Pin bending, from the lever arm across the lug, the gap and the shackle	$M = (P/2)\cdot(t/4 + g + t_s/2)$; $\sigma_b = 32\cdot M/(\pi\cdot d_p^3)$
Utilisation of each mode, its stress over the allowable from the material and design category in force (the sheet adds no design factor of its own)	$u = \sigma/\sigma_{allow}$

Basis: Pinned connections[1](Ch. 3); Shear, bearing and bending stresses[2](Ch. 3).

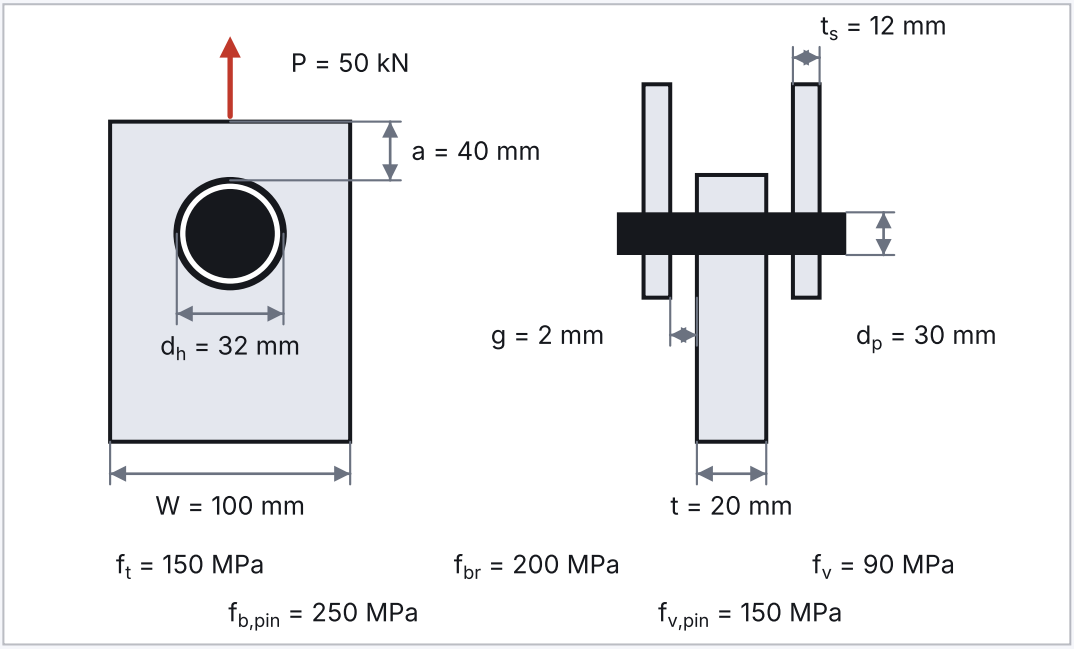


Figure 1 — A lug and its pin, front view and section through the pin: the lug between two cheek plates, the load through the pin. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P	Load on the lug (along the lug)	50	kN	spec: LOAD-SCHEDULE-01	0.01 … 100000 kN
t	Lug thickness	20	mm	drawing: DRG-LL-101 rev A	1 … 500 mm
W	Lug width at the hole	100	mm	drawing: DRG-LL-101 rev A	5 … 2000 mm
d_h	Hole diameter	32	mm	drawing: DRG-LL-101 rev A	2 … 1000 mm
a	Edge distance, hole edge to lug end (load direction)	40	mm	drawing: DRG-LL-101 rev A	1 … 2000 mm
d_p	Pin diameter	30	mm	drawing: DRG-LL-101 rev A	2 … 1000 mm
t_s	Cheek plate thickness (each)	12	mm	drawing: DRG-LL-101 rev A	1 … 500 mm
g	Clearance, lug face to cheek plate (each side)	2	mm	drawing: DRG-LL-101 rev A	0 … 100 mm
f_t	Allowable tensile stress, lug	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
f_{br}	Allowable bearing stress, lug	200	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
f_v	Allowable shear stress, lug	90	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
$f_{b,pin}$	Allowable bending stress, pin	250	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
$f_{v,pin}$	Allowable shear stress, pin	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

3. Assumptions

A1. The load acts along the lug's axis through the hole centre. (No out-of-plane or side load.)
A2. The pin bears uniformly over the lug thickness and over each cheek plate; each cheek plate reacts half the load. (Pin-bending idealisation for a lug between two cheek plates.)
A3. A close-fitting pin ($d_p \geq 0.9 \cdot d_h$). (Uniform-bearing idealisation.)

4. Working

[14] $k_{fit} = d_p / d_h$
$= 30\ \text{mm} / 32\ \text{mm}$
$= 0.9375$
Pin-to-hole ratio

[15] $A_{net} = (W - d_h) * t$
$= (100\ mm - 32\ mm) \times 20\ mm$
$= 1360\ mm^2$
Net section through the hole

[16] $\sigma_t = P / A_{net}$ [1](Ch. 3)
$= 50\ kN / 1360\ mm^2$
$= 36.765\ MPa$
Net-section tension

[17] $A_{br} = d_p * t$
$= 30\ mm \times 20\ mm$
$= 600\ mm^2$
Bearing area

[18] $\sigma_{br} = P / A_{br}$
$= 50\ kN / 600\ mm^2$
$= 83.333\ MPa$
Bearing of the pin on the hole

[19] $A_{to} = 2 * a * t$
$= 2 \times 40\ mm \times 20\ mm$
$= 1600\ mm^2$
Tear-out area (two planes beyond the hole)

[20] $\tau_{to} = P / A_{to}$
$= 50\ kN / 1600\ mm^2$
$= 31.25\ MPa$
Tear-out shear

[21] $A_p = 2 * pi * d_p^2 / 4$
$= 2 \times \pi \times (30\ mm)^2 / 4$
$= 1413.7\ mm^2$
Pin area in double shear

[22] $\tau_p = P / A_p$
$= 50\ kN / 1413.7\ mm^2$
$= 35.368\ MPa$
Pin shear

[23] $M_p = P / 2 * (t / 4 + g + t_s / 2)$
$= 50\ kN / 2 \times (20\ mm / 4 + 2\ mm + 12\ mm / 2)$
$= 325\ N·m$
Pin bending moment (uniform bearing on lug and cheek plates)

[24] $Z_p = pi * d_p^3 / 32$
$= \pi \times (30\ mm)^3 / 32$
$= 2650.7\ mm^3$
Pin section modulus

[25] $\sigma_b = M_p / Z_p$
$= 325\ N·m / 2650.7\ mm^3$
$= 122.61\ MPa$
Pin bending stress

[26] $u_t = \sigma_t / f_t$
$= 36.765\ MPa / 150\ MPa$
$= 0.2451$
Net-section utilisation

[27] $u_{br} = \sigma_{br} / f_{br}$
$= 83.333\ MPa / 200\ MPa$
$= 0.41667$
Bearing utilisation

[28] $u_{to} = \tau_{to} / f_v$
$= 31.25\ \text{MPa} / 90\ \text{MPa}$
$= 0.34722$
Tear-out utilisation
[29] $u_p = \tau_p / f_{v,pin}$
$= 35.368\ \text{MPa} / 150\ \text{MPa}$
$= 0.23579$
Pin shear utilisation
[30] $u_b = \sigma_b / f_{b,pin}$
$= 122.61\ \text{MPa} / 250\ \text{MPa}$
$= 0.49043$
Pin bending utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
σ_t	36.765 MPa	Net-section tension
σ_{br}	83.333 MPa	Bearing stress
τ_{to}	31.25 MPa	Tear-out shear
τ_p	35.368 MPa	Pin shear
M_p	325 N·m	Pin bending moment
σ_b	122.61 MPa	Pin bending stress
u_t	0.2451	Net-section utilisation
u_{br}	0.41667	Bearing utilisation
u_{to}	0.34722	Tear-out utilisation
u_p	0.23579	Pin shear utilisation
u_b	0.49043	Pin bending utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pin bending	$u_b = 0.49043$	$u_b \le 1$	0.49	
PASS	Bearing	$u_{br} = 0.41667$	$u_{br} \le 1$	0.417	[1](Ch. 3)
PASS	Tear-out	$u_{to} = 0.34722$	$u_{to} \le 1$	0.347	[1](Ch. 3)
PASS	Net-section tension	$u_t = 0.2451$	$u_t \le 1$	0.245	[1](Ch. 3)
PASS	Pin double shear	$u_p = 0.23579$	$u_p \le 1$	0.236	

7. Applicability

No warnings.

8. Verification

Template mech.lifting-lug@1.0.1 Content hash 17ed215aa272cf35 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. American Society of Mechanical Engineers. Design of Below-the-Hook Lifting Devices. ASME BTH-1. ASME.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.